- NVIDIA GeForce RTX 3050
- AMD Radeon RX 6500M

For insights on how these GPUs perform in real-world gaming tests, you can refer to my article where I compare GPUs for gaming laptops under Rs 60,000. As a general rule of thumb, the NVIDIA GeForce RTX 3050 stands out as the most powerful GPU, followed by the AMD Radeon RX 6500M. The NVIDIA GTX 1650 and NVIDIA RTX 2050 are closely matched in performance, but more often than not, the GTX 1650 has a slight edge.

Furthermore, the RTX 3050 features DLSS, which can significantly enhance FPS in specific games. However, it's worth noting that DLSS is only compatible with certain games, and its implementation depends on the game developers specifically.

Display

Gaming laptops in the Rs 60,000 price range usually come with a 15.6-inch 1080p display with either 120 or 144Hz refresh rate support. While some people, especially those who play eSports, might notice the difference between 120Hz and 144Hz, for most, the difference will likely be marginal. Another factor to consider in a gaming laptop under Rs 60,000 is peak luminance or brightness. As usual, the higher, the better.

Most gaming laptops under Rs 60,000 have 250 to 300 nits of peak brightness. The last thing related to the display you need to consider is the panel type. Gaming laptops in this price range come with either an IPS or a VA panel. It's recommended to go with the IPS panel for better colour reproduction and viewing angles.

CPU

After the GPU, the CPU is the next crucial component you need to select for a gaming laptop under Rs 60,000. When choosing your CPU, two primary factors to consider are the number of cores and clock speed. Additionally, considering the processor generation is vital. For simplicity, if you're leaning

towards an AMD processor, aim for the AMD Ryzen 5000 series or newer. If you're inclined towards an Intel processor, ensure it's at least an 11th-generation chip or newer.

For those opting for an AMD chip, the primary choices are the 5600H or 5800H. Occasionally, you might come across the Ryzen 5600HS. The "H" signifies high performance, while "HS" indicates an AMD CPU tailored for slim laptops. The 5800H boasts eight performance cores, supporting up to sixteen threads, and a 4.40GHz boost clock speed. Additionally, it also features an 8-core iGPU with a 2000MHz clock speed.

The Ryzen 5800H is the frontrunner, followed by the Ryzen 5600H, which has 6-cores, 12-threads, a marginally lower boost clock speed, and a less potent iGPU. The AMD Ryzen 5600HS rounds out the list. AMD refers to its processors as APUs or Accelerated Processing Units, primarily because these chips combine processing cores and integrated graphics on a single chip. One of the advantages of AMD APUs is their superior multicore performance due to the higher number of performance cores.

On the other hand, if you're considering an Intel CPU, you have the 11th and 12th gen to choose from. Intel's 11th Gen processors predominantly feature high-performance cores, while the 12th Gen divides physical CPU cores between performance and efficiency, functioning as their names suggest. Both the 11th and 12th Gen chips come with Intel Iris Xe iGPU, which is somewhat inferior to the Radeon iGPU in AMD Ryzen chips.

In the gaming laptop under the Rs 60,000 category in India, Intel's Core i5-11320H Processor, with an 8M Cache, can boost up to 4.50 GHz. Intel's Core i5-11400H and i5-11320H both offer a boost clock speed of up to 4.50 GHz, with the former having a 12M Cache and the latter an 8M Cache. Conversely, the Intel Core i5-11260H and i5-12450H can achieve up to 4.40 GHz, each equipped with a 12M Cache. Given their diverse core and thread counts, these processors are engineered for robust performance, making them popular choices among consumers.

RAM

Figuring out how much your budget gaming laptop should have is pretty easy. You need to choose a laptop with at least 16GB RAM. But there are a few things you need to keep in mind. Firstly, make sure that the RAM type is DDR4 and not LPDDR4. The latter type of RAM is soldered to the motherboard and hence not upgradable.

Secondly, always check if the gaming laptop's RAM configuration is running in dual-channel. In other words, it means the laptop has two identical RAM sticks instead of one. Also, RAM speed is also something you should take into account. 3200MHz is the standard RAM speed found in most laptops.

Storage Choice

When selecting the storage option for a laptop, it's crucial to prioritize speed and performance. Traditional hard drives (HDDs) are becoming less common in modern laptops, with solid-state drives (SSDs) taking the lead. SSDs are significantly faster than HDDs, ensuring quicker boot times, faster application launches, and overall improved system responsiveness. If budget allows, opt for a laptop with an